BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2017

CHEMISTRY

Past Paper 2017

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (Multiple Choice Questions (MCQ's))

> Note: Q.1 Choose the correct answer for each from the given options:

Q(i). In determining empirical formula the mole ratio of elements is calculated by dividing the percentage of each element with its:

(a) Atomic number
(b) Atomic mass
(c) Smallest quotient
(d) Formula mass

Q(ii). 64g O2 at S.T.P occupies volume:

(a) 22.4 dm^3
(b) 44.8 dm^3
(c) 11.2 dm^3
(d) 5.6 dm^3

Q(iii). 3.01×10^{20} molecules of glucose ($C_6H_{12}O_6$) have mass in grams:

(a) 0.9
(b) 2.24 m3
(c) 0.0224 m^3
(d) $N.m^2$

Q(iv). In general gas equation when pressure is 101300 Nm^{-2} then volume of gas will be:

(a) 22.4 m3
(b) 2.24 m^3
(c) 0.0224 m^3
(d) 11.2m^3

Q(v). The S.I Unit of viscosity is:

(a) erg. $s.m^2$
(b) $N.s.m^{-2}$
(c) $N.s.m^2$
(d) $N.m^2$

Q(vi). Electron behaves as a wave is the statement of:

(a) Niels Bohr
(b) Rutherford
(c) Heisenberg
(d) De Broglie

Q(vii). The maximum number of electrons which can be accommodated in an energy sub level is determined by formula:

(a) $2(l + 1)$
(b) $2(2l + 1)$
(c) $2(2l + 2)$
(d) $(n + l)$

Q(viii). For the removal of valence electron from Na atom the value of energy required in KJ/moles is:

(a) 395
(b) 595
(c) 495
(d) 594

Q(ix). The molecule which has two partially filled 2p_y and 2p_z orbitals is:

(a) N_2
(b) H_2
(c) Cl_2
(d) O_2

Q(x). The bond angle in CH_4 molecule is equal to:

(a) 109.5°
(b) 104.5°
(c) 107.5°
(d) 90°

Q(xi). In a reversible reaction when there is a decrease in the volume on the product side the relation is expressed as:

(a) $K_p = K_c$
(b) $K_p > K_c$
(c) $K_p < K_c$
(d) None of these

Q(xii). Ionic product of dissolved ions is called as:

(a) Common Ion Effect
(b) Solubility Product
(c) Normality
(d) Molarity

Q(xiii). The ability of an ion to form hydrate depends upon:

(a) Molecular mass
(b) Atomic mass
(c) Charge density
(d) Nature of ion

Q(xiv). Hydrolysis does not occur in the solution (aq.) of:

(a) NaOH
(b) [illegible in source scan]Cl
(c) NaCl
(d) NH_4Cl

Q(xv). In super oxide the O.N. of oxygen is:

(a) 2^-
(b) 1^-
(c) $-1/2$
(d) $1/2$

Q(xvi). Oxidation number of Mn in MnO_4^- is:

(a) 1^+
(b) 3^+
(c) 5^+
(d) 7^+

Q(xvii). The sum of exponents of concentration is called as:

(a) Rate of reaction
(b) Order of reaction
(c) Molecularity of reaction
(d) Fast reaction

Section-B (Short Answer)

Maximum Marks: 40

> Note: Answer any EIGHT of the following questions. Each question carries 05 marks.

Q2. (Marks 05) Define the following with examples: (a) Error and Deviation (b) Amorphous Solids

Q3. (Marks 05) What is Avogadro's number? Calculate number of atoms in 12 g of Mg (Atomic mass = 24 a.m.u)

Q4. (Marks 05) Explain the main postulates of kinetic molecules theory of gases.

Q5. (Marks 05) What volume does 800 cm3 sample of a gas at 350 torr occupy when pressure is changed to 3 atom?

Q6. (Marks 05) Discuss with examples Tetragonal and Trigonal Crystal System.

Q7. (Marks 05) Define Graham's law of Diffusion. Compare rate of diffusion of pair of gases. (i) H_2 and D_2 (b) CH_4 and H_2

Q8. (Marks 05) Explain the origin of X-rays with the relationship between their wavelength and nuclear charge in the atoms from which they originate.

Q9. (Marks 05) What is meant by the term Dipole Moment? Discuss its significance.

Q10. (Marks 05) In a certain process 800 J of heat is absorbed by a system, while 300J of work is done on the system. What is the change in the internal energy?

Q11. (Marks 05) What is essential difference between hydration and hydration and hydrolysis?

Section-C (Descriptive Answer)

Maximum Marks: 28

> Note: Answer any TWO of the following questions. Each question carries 14 (7+7) marks.

Q12.

(a) [07] What is meant by Solubility Product and Common Ion Effect? Discuss the application of Common Ion Effect in the qualitative analysis.

(b) [07] Why Beryllium chloride has a linear shape where a water molecule has angular shape? Discuss in terms of electron repulsion model.

Q13.

(a) [07] Explain hybridization, How sp^3 hybrid orbital differ from s-and p-orbitals?

(b) [07] What is meant by Equilibrium Constant and how it is determined experimentally?

Q14.

(a) [07] Balance the following equations by Ion Electron method. (i) $Cr_2O_7^{2-} + H^+ + I_2 \rightarrow Cr^{3+} + IO_3^- + H_2O$ (ii) $MnO_4^- + SO_3^{2-} + OH^- \rightarrow Mn^{2+} + SO_4^{2-}$

(b) [07] Write short notes on any TWO of the following: (i) Radioactivity (ii) Characteristics of Ionic and Covalent Compounds (iii) Determination of Rate of Reaction.